SDN Questions

8 questions | Original English questions and source answers

Question 1

What are two benefits of controller-based networking compared to traditional networking? (Choose two)

Options

A. controller-based increases network bandwidth usage, while traditional lightens the load on the network.

B. controller-based reduces network configuration complexity, while traditional increases the potential for errors

C. controller-based provides centralization of key IT functions, while traditional requires distributes management function

D. controller-based allows for fewer network failure, while traditional increases failure rates

E. controller-based inflates software costs, while traditional decreases individual licensing costs

Answer

B. controller-based reduces network configuration complexity, while traditional increases the potential for errors

C. controller-based provides centralization of key IT functions, while traditional requires distributes management function

Source: CCNA Training » SDN Quiz.pdf | Question 1

Question 2

In software defined architectures, which plane is distributed and responsible for traffic forwarding?

Options

A. management plane

B. control plane

C. policy plane

D. data plane

Answer

D. data plane

Source explanation

The Open Networking Foundation identifies three main parts of the Software-defined networking (SDN): Application layer; Control layer and Infrastructure layer. SDN separates a router’s control plane from the forwarding plane. The control plane makes routing decisions. The data plane forwards data (packets) through the router. With SDN routing, decisions are made remotely instead of on each individual router.

Source: CCNA Training » SDN Quiz.pdf | Question 2

Question 3

What software defined architecture plane assists network devices with making packet forwarding decisions by providing Layer 2 reachability and Layer 3 routing information?

Options

A. policy plane

B. management plane

C. data plane

D. control plane

Answer

D. control plane

Source explanation

Control Plane Function
In its simplest form, the control plane provides layer-2 MAC reachability and layer-3 routing information to network devices that require this information to make packet forwarding decisions. In the case of firewalls, the control plane would include stateful flow information for inspection. Control plane functionality can implemented as follows:
+ Distributed — Conventional routers and switches operate using distributed protocols for control, i.e. where each device makes its own decisions about what to do, and communicate relevant information to other devices for input into their decision making process. For example, the Spanning Tree Protocol (STP), Fabric Path, and routing protocols such as IS-IS and BGP provide distributed control of packet forwarding functionality to networking devices.
+ Centralized — In this case, a centralized controller provides the necessary information for a network element to make a decision. For example, these controller(s) instruct networking devices on where to forward packets by explicitly programming their MAC and FIBs.
Reference: https://www.cisco.com/c/en/us/td/docs/solutions/Enterprise/Data_Center/VMDC/SDN/SDN.html

Source: CCNA Training » SDN Quiz.pdf | Question 3

Question 4

Which statement correctly compares traditional networks and controller-based networks?

Options

A. Traditional and controller-based networks abstract policies from device configurations

B. Only traditional networks offer a centralized control plane

C. Only traditional networks natively support centralized management

D. Only controller-based networks decouple the control plane and the data plane

Answer

D. Only controller-based networks decouple the control plane and the data plane

Source explanation

Most traditional devices use a distributed architecture, in which each control plane is resided in a networking device. Therefore they need to communicate with each other via messages to work correctly.
In contrast to distributed architecture, centralized (or controller-based) architectures centralizes the control of networking devices into one device, called SDN controller -> Answer 'Only controller-based networks decouple the control plane and the data plane' is correct.

Source: CCNA Training » SDN Quiz.pdf | Question 4

Question 5

Which statement about the Cisco ACI fabric is most accurate?

Options

A. An APIC is a cluster of at least three APIC controllers, providing a single point of management without a single point of failure.

B. The fabric header carries the EPG from the egress to the ingress leaf switch.

C. The APIC is able to enforce security by inserting itself into the data path.

D. The spine switch rewrites the EPG from ingress to egress when it performs the forwarding proxy function

Answer

A. An APIC is a cluster of at least three APIC controllers, providing a single point of management without a single point of failure.

Source: CCNA Training » SDN Quiz.pdf | Question 5

Question 6

What are two southbound APIs? (Choose two)

Options

A. Open Flow

B. NETCONF

C. Thrift

D. DSC

E. CORBA

Answer

A. Open Flow

B. NETCONF

Source explanation

OpenFlow is a well-known southbound API. OpenFlow defines the way the SDN Controller should interact with the forwarding plane to make adjustments to the network, so it can better adapt to changing business requirements.
The Network Configuration Protocol (NetConf) uses Extensible Markup Language (XML) to install, manipulate and delete configuration to network devices.
Other southbound APIs are:
+ onePK: a Cisco proprietary SBI to inspect or modify the network element configuration without hardware upgrades.
+ OpFlex: an open-standard, distributed control system. It send “summary policy” to network elements.

Source: CCNA Training » SDN Quiz.pdf | Question 6

Question 7

What are two characteristics of a controller-based network? (Choose two)

Options

A. It uses northbound and southbound APIs to communicate between architectural layers

B. It decentralizes the control plane, which allows each device to make its own forwarding decisions

C. The administrator can make configuration updates from the CLI

D. It moves the control plane to a central point

E. It uses Telnet to report system issues.

Answer

A. It uses northbound and southbound APIs to communicate between architectural layers

D. It moves the control plane to a central point

Source: CCNA Training » SDN Quiz.pdf | Question 7

Question 8

Which API is used in controller-based architectures to interact with edge devices?

Options

A. underlay

B. southbound

C. northbound

D. overlay

Answer

B. southbound

Source explanation

The Southbound API is used to communicate with network devices.

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